# Neuroanatomy

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VIP ACADEMY

# ***Spinal cord***

**1- In adults, the spinal cord ends at: *(2019 - 2020)*.**

a- Lower border of L1.

b- Lower border of L3.

c- Lower border of S1.

d- Lower border of L3.

**2- A patient came to emergency room presenting with headache; fever, nausea and stiffness of the neck. He was diagnosed as meningitis. What is the name of the procedure by which a special needle is inserted into the subarachnoid space to get sample of CSF to confirm the diagnosis? *(2018 – 2017 - 2016)*.**

a- Conus medullaris.

b- Lumbar cistern.

c- Lumbar puncture.

d- Cauda equine.

**3- Cauda equina is: *(2018)*.**

a- The lower segments of the spinal cord.

b- The roots of lower lumbar, sacral and coccygeal nerves.

c- The spinal nerves of the conus medullaris.

d- The spinal nerves of the lower lumbar and sacral segments.

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**4- The spinal cord has ..... select the best statement *(DEP BOOK)*.**

a- An outer covering of gray matter and an inner core of white matter.

b- Anterior and posterior roots of a single spinal nerve attached to a single segment.

c- Cells in the posterior gray horn give rise to efferent fibers that supply skeletal muscle.

d- A central canal that is situated in the white commissure.

**5- If we need to insert a needle into the subarachnoid space to sample CSF, where is a relatively safe point for needle insertion? *(DEP BOOK)*.**

a- Between C-7 and T-1.

b- Between L-3 and L-4.

c- Between L-1 and L-2.

d- Between T-12 and L-1.

**6- In newborns, conus medullaris lies opposite the: *(2021 – IPKA - 2023)*.**

a- 10$^{th}$ thoracic vertebra.

b- 1$^{st}$ lumbar vertebra.

c- 3$^{rd}$ lumbar vertebra.

d- 5$^{th}$ lumbar vertebra.

**7- One of the following arteries is not supplying spinal cord: *(2021)*.**

a- Radicular branches of vertebral artery.

b- Posterior spinal artery.

c- Medial spinal artery.

d- Anterior spinal artery.

**8- The true statement about the spinal cord is: *(2020 - 2019)*.**

a- The spinal dura end at the level of S4.

b- It has four enlargements.

c- The filum terminal is a fold of pia below the termination of spinal cord attached to the. coccyx.

d- The largest radicular artery is that of S2.

**9- Cauda equine, choose one true statement: *(2020)*.**

a- Formed of all thoracic and lumber nerves.

b- Occupy lower part of spinal canal.

c- Surrounded by the three layers of the meninges.

d- Pass through the sacrum to end at S2.

**10- The lower end of spinal cord is called: *(2019)*.**

a- Cauda equina.

b- Denticulate ligament.

c- Conus medullaris.

d- Ganglion impar.

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**11- Filum terminal is attached to: *(IPKA)*.**

a- S3.

b- S4.

c- Foramen magnum.

d- Back of coccyx.

**12- Lumbar puncture can be done safely: *(IPKA)*.**

a- Between L1 and L2 vertebrae.

b- Between L3 and L4 vertebrae.

c- Between S3 and S4 vertebrae.

d- Between T3 and T4 vertebrae.

**13- Anterior spinal artery supplies: *(IPKA)*.**

a- Posterior 2/3 of spinal cord.

b- Lateral 2/3 of spinal cord.

c- Posterior 1/3 of spinal cord.

d- Anterior 2/3 of spinal cord.

**14- Regarding the spinal cord, select the true statement: *(2023)*.**

a- The lumbosacral enlargement called conus medullaris.

b- Denticulate ligament are lateral extension of arachnoid.

c- The adult spinal cord terminates at intervertebral disc between L2 and L3.

d- The filum terminale attach the lower end of spinal cord with the back of coccyx.

**15- The arterial supply of the lumber enlargement of the spinal cord: *(2024)*.**

a- The arteria radicularis magna.

b- The anterior spinal artery.

c- The posterior spinal arteries.

d- The basilar artery.

**16- Concerning spinal nerve *C-8,* which of the following is true? *(DEP BOOK)*.**

a- C-8 enters/exits between vertebrae C6 and C7.

b- C-8 enters/exits between vertebrae C7 and C8.

c- C-8 enters/exits between vertebrae C7 and C8.

d- C-8 enters/exits between vertebrae C7 and T1.

**17- In adults, the spinal cord normally ends at: *(DEP BOOK)*.**

a- Lower border of L1.

b- Lower border of L3.

c- Lower border of S1.

d- Lower border of L5.

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**18- Regarding the spinal meninges, one of the following is true: *(DEP BOOK)*.**

a- The spinal arachnoid matter ends at the level of L2.

b- The spinal arteries lie in the epidural space.

c- The lower part of the spinal dura forms a sheath around the filum terminale.

d- The lumbar cistern extends from L1 to S3.

**19- Regarding the blood supply of the spinal cord, one of the following is true: *(DEP BOOK)*.**

a- The posterior spinal artery supplies the anterior and lateral funiculi of the white matter.

b- The arteria radicularis magna supplies the cervical enlargement.

c- The internal vertebral venous plexus lies in the subdural space.

d- The thoracic region of the spinal cord is more liable.

**20- Regarding the meninges of the spinal cord, one of the following is true: *(DEP BOOK)*.**

a- The dura mater ends at the level of the third piece of sacrum.

b- The arachnoid mater ends at the level of the forth piece of sacrum.

c- The pia mater forms ligamentum denticulatum.

d- The subdural space contains the vertebral venous plexuses.

**21- Regarding the spinal cord, one of the following is true: *(DEP BOOK)*.**

a- The spinal dura ends at the level of S4.

b- It is drained by spinal and vertebral venous plexuses.

c- It has three enlargements.

d- The largest radicular artery is that of S2.

**22- The lower end of spinal cord is called: *(DEP BOOK)*.**

a- Cauda equina.

b- Denticulate ligament.

c- Conus medullaris.

d- Ganglion ispar.

**23- Substantia gelatinosa of Rolandi forms the second order neurons in the pathway of: *(DEP BOOK)*.**

a- Pain only.

b- Temperature only.

c- Pain and temperature.

d- Simple touch.

**24- The posterior funiculus of the spinal cord contains: *(DEP BOOK)*.**

a- Long descending fibers.

b- Long ascending fibers.

c- Both long ascending and descending fibers.

d- Associative fibers.

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**25- Regarding the pyramidal tract choose the correct answer: *(DEP BOOK)*.**

a- Consists of many neurons.

b- It arises only from the pyramidal cells of area 4 of the cerebral cortex.

c- About 15 % of its fibers normally form the pyramidal decussation.

d- It has a corticospinal and corticonuclear parts.

**26- Regarding Brown Sequard syndrome; choose one best answer: *(DEP BOOK)*.**

a- It is associated with loss of reflexes at the level of the lesion.

b- It is usually associated with injury to nerve roots entering the segments below the lesion.

c- There is pyramidal tract lesion on the opposite side and proprioceptive lesion on the same side.

d- It is associated with loss of reflexes below the lesion.

**27- Regarding the blood supply to the spinal cord, select the correct statement: *(DEP BOOK)*.**

a- Anterior 2/3 of the spinal cord is supplied by the posterior spinal arteries.

b- Anterior spinal arteries are reinforced in the cervical region by branches from the vertebral arteries.

c- There are two anterior spinal arteries.

d- The cervical region of the spinal cord is predisposed to vascular insufficiency.

**28- Regarding the spinal meninges select the correct statement: *(DEP BOOK)*.**

a- The denticulate ligament attaches pia to the arachnoid mater.

b- The pia mater is attached to the foramen magnum.

c- The arachnoid mater ends at L2 level.

d- The dura mater sends sheaths around the spinal nerve roots.

**29- Regarding the spinal meninges, select the true statement: *(2019)*.**

a- The denticulate ligaments are thickening of the dura mater.

b- The dura mater continues below as the film terminale.

c-The pia matter is closely adherent to the spinal cord.

d- The dura mater ends at the level of the fourth sacral vertebra.

**30- In performing a lumbar puncture for CSF collection, the needle Is Inserted Into: *(2021 – 2023 - IPKA)*.**

a- Subdural space.

b- Epidural space.

c- Subarachnoid space.

d- Extradural space.

## ***Brain stem***

**1- The pyramidal tract lies in the midbrain in one of the following parts: *(2018)*.**

a- Medial 1/5 of crus cerebri.

b- Lateral 1/5 of crus cerebri.

c- Intermediate of 3/5 of crus cerebri.

d- Dorsal to crus cerebri.

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**2- A 50-year-old patient came to outpatient clinic. On examination, the physician found that there was a hemiplegia on the left side and medial squint on the right side. What are structures most likely affected? *(2018)*.**

a- Left pyramidal tract and left abducent nerve.

b- Right pyramidal tract and right abducent nerve.

c- Left pyramidal tract and left oculomotor nerve.

d- Right pyramidal tract and right oculomotor nerve.

**3- The medial medullary syndrome (Dejerine syndrome) is characterized by: *(2018)*.**

a- Ipsilateral hemiplegia and hypoglossal paralysis.

b- Ipsilateral hemiplegia with contralateral hypoglossal paralysis.

c- Ipsilateral hypoglossal paralysis with contralateral hemiplegia.

d- Contralateral hemiplegia and hypoglossal paralysis.

**4- Patient came to ER with LMNL of the hypoglossal nerve on the right side and UMNL hemiplegia on the left side. Where is the site of the lesion? *(2017)*.**

a- Motor area 4.

b- Midbrain.

c- Pons.

d- Medulla oblongata.

**5- The pyramidal tract fibers supply the following cranial nerve nucleus: *(2017)*.**
a- Olfactory.
b- Optic.
c- Trigeminal.
d- Vestibulocochlear.
**6- The following artery shares in the supply of the medulla: *(2016)*.**
a- Posterior inferior cerebellar.
b- Superior cerebellar.
c- Posterior cerebral.
d- Middle cerebral.
**7- Which of the following cranial nerves exits the posterior aspect of the brainstem? One of the following cranial nerves is attached to the posterior aspect of the brain stem: *(DEP BOOK - 2024)*.**
a- Abducens (VI).
b- Hypoglossal (XII).
c- Trigeminal (V).
d-Trochlear (IV).
e- Oculomotor (III).
f- Vagus (X).
**8- 50-year-old patient came to outpatient clinic. On examination, the physician found that there was a hemiplegia on the left side and medial squint on the right side. What are structures most likely affected? *(2017)*.**
a- Left pyramidal tract and left abducent nerve.
b- Right pyramidal tract and right abducent nerve.
c- Left pyramidal tract and left oculomotor nerve.
d- Right pyramidal tract and right oculomotor nerve.
**9- In case of right corticonuclear tract injury above the level of the motor nucleus of. facial nerve, which side of the face is affected? *(2017)*.**
a· Upper ½ right side.
b- Lower ½ right side.
c- Upper½ left side.
d- Lower ½ left side.
**10- Facial colliculus overlies *(2015)*.**
a- Abducent nucleus.
b- Facial motor nucleus.
c- Hypoglossal nucleus.
d- Dorsal nucleus of X.
e- Cochlear nuclei.
**11- Concerning the olive, select the correct statement: *(DEP BOOK)*.**
a- Lies between the median and anterolateral fissures.
b- Is related posteriorly to the rootlets of hypoglossal nerve.
c- Is related posteriorly to the rootlets of 9, 10 and 11 cranial nerves.
d- Is produced by the superior olivary nucleus.

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**12- Regarding the olive, one of the following is incorrect: *(2021 - IPKA)*.**
a- Lies on the lateral surface of medulla oblongata.
b- Is related anteriorly to the rootlets of hypoglossal nerve.
c- Is related posteriorly to the rootlets of 9, 10 and 11 cranial nerves.
d- Is produced by the superior olivary nucleus.
**13- The facial colliculus is formed by the: *(2021 – IPKA - 2023)*.**
a- Facial nerve fibers and abducent nucleus.
b- Facial nerve fibers only.
c- Facial nucleus only.
d- Facial nucleus and abducent nerve fibers.
**14- Concerning the midbrain, select the wrong answer: *(2021)*.**
a- The trochlear nerve emerges from its ventral surface.
b- Is traversed by the aqueduct of Sylvius.
c- Lies between the pons below and diencephalon above.
d- Its tectum is connected to the metathalamus by 2 brachia on each side.
**15- The tectum is a part of: *(2020)*.**
a- Diencephalon.
b- Midbrain.
c- Medulla oblongata.
d- Pons.
**16- Facial colliculus is seen in: *(2020)*.**
a- Pons.
b- Medulla.
c- Midbrain.
d- Interpeduncular fossa.
**17- About the medulla oblongata, choose the correct answer: *(2020 - 2019)*.**
a- The olive is produced by the superior olivary nucleus.
b- The hypoglossal nerve is attached to the posterolateral sulcus.
c- The vagus nerve is attached to the anterolateral sulcus.
d- The pyramid is an elongated elevation produced by corticospinal tract.
**18- The back of the closed medulla shows two elevations: *(2019)*.**
a- Gracile and cuneate tubercles.
b- Pyramid and olive.
c- Olive and inferior cerebellar peduncle.
d- Facial colliculus and vestibular area.
**19- The cranial nerves related to the pons are: *(2019)*.**
a- 7,8,9.
b- 8, 9, 10.
c- 5, 6, 7,8.
d- 2, 3, 4.

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**20- The inferior fovea is present in: *(IPKA)*.**
a- Back of pons.
b- Closed medulla.
c- Open medulla.
d- Back of midbrain.
**21- One of the followings is present in the posterior aspect of open medulla: *(IPKA)*.**
a- Gracile tubercle.
b- Cuneate tubercle.
c- Hypoglossal trigone.
d- Superior fovea.
**22- Olive of the medulla is produced by: *(IPKA)*.**
a- Cortico-bulbar tract.
b- Cortico-spinal tract.
c- Inferior olivary nucleus.
d- Hypoglossal nucleus.
**23- Hypoglossal nerve emerges along: *(IPKA)*.**
a- Posterolateral sulcus.
b- Anterior median fissure.
c- Posterior median fissure.
d- Anterolateral sulcus.
**24- A cranial nerve emerges from Cerebellopontine angle: *(IPKA)*.**
a- Abducent nerve.
b- Trigeminal nerve.
c- Vagus nerve.
d- Facial.
e- Vagal nucleus.
**25- The posterior part of the midbrain is formed by: *(2023)*.**
a- Tectum.
b- Tegmentum.
c- Tapetum .
d- Superior cerebellar peduncle.
**26- The following structure wind around the lateral surface of midbrain: *(2023)*.**
a- Optic tract.
b- Anterior cerebral artery.
c- Abducent nerve.
d- Fornix.
**27- The anterolateral sulcus of medulla between pyramid and olive transmits: *(2023)*.**
a- Hypoglossal nerve.
b- Accessory nerve.
c- Vagus nerve.
d- Glossopharyngeal nerve.

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**28- Regarding the olive one of the following is incorrect: *(2023)*.**
a-Lies on the lateral surface of medulla oblongata.
b- Is related anteriorly to the rootlets of hypoglossal nerve.
c- Is related posteriorly to the rootlets of 9, 10 and 11 cranial nerves.
d- Is produced by the superior olivary nucleus.
**29- The posterior surface of mid brain is represented by: *(2024)*.**
a- Upper part of 4th ventricle.
b- Lower part of 4th ventricle.
c- Cerebral peduncle.
d- Corpora quadrigemina.
**30- One of the following cranial nerves contains sensory nerves fibers: *(DEP BOOK)*.**
a- Occulomotor nerve.
b- Trochlear nerve.
c- Trigeminal nerve.
d- Abducent nerve.
**31- One of the following cranial nerves is attached to the posterior aspect of the brain stem: *(DEP BOOK)*.**
a- Abducent nerve.
b- Vagus nerve.
c- Oculomotor nerve.
d- Trochlear nerve.
**32- In lateral medullary syndrome (Wallenberg) the lesion is in one of the following arteries: *(DEP BOOK)*.**
a- Posterior inferior cerebellar artery.
b- Anterior inferior cerebellar artery.
c- Superior cerebellar artery.
d- Vertebral artery.
**33- Nucleus in brain stem common to IX, X and XI cranial nerves: *(DEP BOOK)*.**
a- Nucleus solitarius.
b- Nucleus ambiguous.
c- Dentate nucleus.
d- Red nucleus.
**34- Regarding the mid brain, one of the following is true: *(DEP BOOK)*.**
a- The cerebral aqueduct of the midbrain connects the 3rd and lateral ventricles.
b- It occupies the tentorial notch of the tentorium cerebelli.
c- The facial nerve curves forwards around its lateral surface.
d- The oculomotor nerve is attached to the tectum above the inferior colliculus.
**35- The cerebellopontine angle contains one the following structures: *(DEP BOOK)*.**
a- The posterior inferior cerebellar artery.
b- The trigeminal nerve.
c- The facial nerve.
d- The vagus nerve.

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**36- The lateral surface of the cerebral peduncle is related to the following structure: *(DEP BOOK)*.**
a- Optic nerve.
b- Anterior cerebellar artery.
c- Posterior cerebral artery.
d- Middle cerebral artery.
**37- The following nucleus is present in the medulla: *(DEP BOOK)*.**
a- The ventral cochlear nucleus.
b- The facial nucleus.
c- The superior vestibular nucleus.
d- The inferior olivary nucleus.
**38- Regarding medulla oblongata, one of the following is true: *(DEP BOOK)*.**
a- Decussation of pyramids represents the crossing of the corticospinal fibers.
b- Vagus nerve emerges between the pyramid and olive.
c- Hypoglossal nerve emerges between olive and inferior cerebellar peduncle.
d- Gracile and cuneate tubercles are located posteriorly in its upper part.
**39- Regarding the midbrain, one of the following is true: *(DEP BOOK)*.**
a- It is traversed by interventricular foramen.
b- Four colliculi lie on its dorsal surface.
c- Superior colliculi are centers for auditory pathway.
d- Inferior colliculi are part of visual pathway.
**40- One of the followings structures belongs to the auditory pathway: (*DEP BOOK)*.**
a- Lateral leminiscus.
b- lateral geniculate body.
c- superior colliculus.
d- Medial leminiscus.
**41- A tumor of the pituitary gland, by compressing the optic chiasma, can cause: *(DEP BOOK)*.**
a- Complete blindness.
b- Bitemporal hemianopia.
c- Binasal hemianopia.
d- Left temporal and right nasal hemianopia.
**42- The pyramidal tract lies in the midbrain in one of the following parts: *(DEP BOOK)*.**
a- Medial 1/5 of crus cerebri.
b- Lateral 1/5 of crus cerebri.
c- Intermediate of 3/5 of crus cerebri.
d- Dorsal to crus cerebri.
**43- Select the correct statement: *(DEP BOOK)*.**
a- The spinal cord has one anterior lateral sulcus.
b- The medulla has anterior median fissure.
c- The lower part of the medulla does not have a central canal.
d- The cerebellum is connected to midbrain by middle cerebellar peduncle.

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**44- Regarding the brain stem select the correct statement: *(DEP BOOK)*.**
a- The pyramidal decussation lies in the lower part of the pons.
b- The central canal of the medulla opens into the 3rd ventricle at a point called the obex.
c- The pons is connected to the cerebellum by superior cerebellar peduncle.
d- The tectum is the posterior part of the midbrain.
**45- The rootlets of the vagus nerve are emerging between (*2016*).**
a- Pyramid and olive.
b- Pyramid and pons.
c- Olive and inferior cerebellar peduncle.
d- Gracile and cuneate tubercles.
**46- The superior brachium connects the superior colliculus with: (*2023*).**
a- Hypothalamus.
b- Lateral geniculate body.
c- Uncus.
d- Mammillary body.
**47- Regarding the medulla oblongata, one of the following is true; (*DEP BOOK*).**
a- The olive is caused by the superior olivary nucleus.
b- The hypoglossal nerve is attached to the posterolateral fissure.
c- The vagus nerve is attached to the anterolateral fissure.
d- The pyramidal decussation obliterates the lower part of the anterior median fissure.
**48- A 68-year-old man came to the out-patient clinic with forward tilting of the trunk and short steps. The examination revealed rigidity and trembling of the head and extremities. The case was diagnosed as Parkinson's disease. Which area of the midbrain is suspected to be affected? *(2016)*.**
a- Crus cerebri.
b- Substantia nigra.
c- Tegmentum.
d- Tectum.

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## ***Cerebellum***

**1- Regarding the paleocerebellum, choose the correct statement: *(2018)*.**
a- It is connected to the vestibular system.
b- It is connected to the spinal cord.
c- It is connected to the cerebral cortex.
d- It is concerned with equilibrium.
**2- The part of the cerebellum concerned with planning and programming movement is: *(2017)*.**
a- Spinocerebellar.
b- Vestibulocerebellum.
c- Archicerebellum.
d- Neocerebellum.

**3- The part of the cerebellum concerned with balance is: *(2016)*.**
a- Paleocerebellum.
b- Neocerebellum.
c- Flocculonodular.
d- Lateral zone area.
**4- Concerning the cerebellum, select the correct statement: *(DEP BOOK)*.**
a- It shares in the formation of the third ventricle.
b- It lies in the middle cranial fossa.
c- It has three peduncles connecting it to the midbrain.
d- Its largest nucleus is the dentate nucleus.
**5- The neocerebellum projects to the motor cortex via: *(DEP BOOK)*.**
a- Anterior thalamic nucleus.
b- Ventral lateral nucleus .
c-Lateral dorsal nucleus.
d-Lateral posterior nucleus.
**6- One of the following regarding the cerebellum is not true: *(2021)*.**
a- Lies in the posterior cranial fossa.
b- Is divided into three lobes.
c- Is separated from the posterior part of the cerebrum by the tentorium cerebelli.
d- Is a part of the brainstem.
**7- The superior cerebellar peduncle connects the cerebellum with: *(2020)*.**
a- Pons.
b- Medulla.
c- Midbrain.
d- Inferior olivary nucleus.
**8- The medulla is connected to the cerebellum through: *(IPKA)*.**
a- Superior cerebellar peduncle.
b- Middle cerebellar peduncle.
c- Inferior cerebellar peduncle.
d- Cerebral peduncle.
**9- Concerning the cerebellum, select the true statement: *(2023)*.**
a- It lies in the middle cranial fossa .
b- Its upper surface is covered by falx cerebri.
c- Its superior vermis has a nodule and uvula only.
d- It's supplied by superior cerebellar artery.
**10- The fissure that separates the flocculonodular lobe from posterior lobe of the cerebellum: *(2024)*.**
a- Primary fissure.
b- Horizontal fissure.
c- Posterolateral fissure.
d- Lateral fissure.

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**11) Regarding the cerebellum, choose one correct answer: *(DEP BOOK)*.**
a- Middle cerebellar peduncle connects it to medulla oblongata.
b- Spinocerebellum regulates muscle tone.
c- Neocerebellum is concerned with equilibrium.
d- Paleocerebellum is concerned with planning and programming· movement.
**12- Regarding the cerebellum, one of the following is true: *(DEP BOOK)*.**
a- The neocerebellum connected to the cerebral cortex.
b- The primary fissure present on its lower surface.
c- It consists of two lobes.
d- The superior vermis has nodule, uvula & pyramid.
**13- Regarding the cerebellum, one of the following is true: *(DEP BOOK)*.**
a- The neocerebellum consists of the flocculonodular lobe.
b- The posterior lobe lies in front of the primary fissure.
c- It lies below the tentorium cerebelli.
d- The superior vermis is found at the bottom of the vallecula.
**14- One of the following is part of inferior vermis: *(DEP BOOK)*.**
a- Culmen.
b- Declive.
c- Folium vermis.
d- Uvula.
**15- Regarding the cerebellum, one of the following is true: *(DEP BOOK)*.**
a- It consists of two hemispheres and diencephalon.
b- Its inferior surface shows a deep groove, the vallecula cerebelli.
c- Its inferior surface shows the fissure prima.
d- Grey matter is found only in its cortex.
**16- Concerning the cerebellum; choose the most correct answer: *(DEP BOOK)*.**
a- Related to the mastoid air cells.
b- Related to the cavernous sinus.
c- It is a part of forebrain.
d- Is related to the 3rd ventricle.
**17- Select the correct statement about the cerebellum: *(DEP BOOK)*.**
a- The cerebellum has two surfaces; upper and lower.
b- The cerebellum has two lobes; anterior and posterior.
c- The anterior notch is occupied by the falx cerebrii.
d- The flocculus is a small part of the cerebral hemisphere.

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# ***Fourth ventricle***

**1- The central canal of the spinal cord opens out into the: *(2019)*.**
a- Subarachnoid space.
b- Lumbar cistern.
c- Third ventricle.
d- Fourth ventricle.
**2- The choroid plexus of the 4th ventricle is derived from the following artery: *(2017)*.**
a- Posterior inferior cerebellar.
b- Anterior inferior cerebellar.
c- Superior cerebellar.
d- Middle cerebral.
**3- The MRI of an 11-year-old boy shows a tumor in the pontine portion of the fourth ventricle. The rostral edge of which of the following structures represents the border between the medullary and pontine parts of the fourth ventricle? *(DEP BOOK)*.**
a- Facial colliculus.
b- Hypoglossal trigone.
c- Medial eminence.
d- Stria medullaris.
**4- Concerning the 4th ventricle, select the correct statement: *(DEP BOOK)*.**
a- Closed medulla oblongata shares in its floor.
b- Is roofed by the superior velum only.
c- Communicates with the lateral ventricle through aqueduct of Sylvius.
d- Communicates with the subarachnoid space through one median and 2 lateral foramina.
**5- One of the following cranial nerve nuclei is not present in the floor of the 4th ventricle: *(2021)*.**
a- Abducent nucleus.
b- Oculomotor nucleus.
c- Dorsal motor nucleus of the vagus.
d- Hypoglossal nucleus.
**6- Foramen of Luschka connects the: (*2021)*.**
a- Lateral ventricle with the third ventricle.
b- Two lateral ventricles.
c- Third ventricle with the fourth ventricle.
d- Fourth ventricle with the subarachnoid space.
**7- Concerning the fourth ventricle; choose the correct answer: *(2019 – DEP BOOK)*.**
a- Its floor is formed by the closed medulla and pons.
b- Its roof is formed by the superior and inferior medullary vela.
c- Its choroids plexus is present in the floor of the ventricle.
d- It has a communication with the lateral ventricle.

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**8- The upper angle of 4th ventricle communicates it with: *(IPKA)*.**
a- Cerebral aqueduct of midbrain.
b- Central canal of medulla oblongata.
c- Foramen of Magendi.
d- Foramen of Monro.
**9- A foramen that connects the 4th ventricle with the subarachnoid space: *(IPKA)*.**
a- Central canal of mid brain.
b- Foramen of Magendi.
c- Interventricular foramen.
d- Central canal of medulla.
**10- The roof of the 4th ventricle is formed by: *(2023)*.**
a- Cerebellum.
b- Corpus callosum.
c- Tectum.
d- Thalamus.
**11- Which ventricle lies between the pons and cerebellum? (*2024)*.**
a- Lateral ventricle.
b- Third ventricle.
c- Fourth ventricle.
d- Fifth ventricle.
**12- Concerning the fourth ventricle; choose the most correct answer. (*DEP BOOK)*.**
a- Its floor is formed by the closed medulla and pons.
b- Its roof is formed by the superior and inferior medullary vela.
c- Its choroids plexus is present in the floor of the ventricle.
d- It has a communication with the lateral ventricle.
**13- Regarding the fourth ventricle, mark ONE true statement. *(DEP BOOK)*.**
a- It is the cavity of the telencephalon.
b- It lies dorsal to cerebellum.
c- Its choroid plexus hangs from its roof.
d- Superior medullary velum is a layer of ependyma

VIP ACADEMY

## ***Sulci & gyri***

**1- One of the following is incorrect concerning the insula: *(2021)*.**
a- It is a conical mass of grey mater lying in the depth of the lateral sulcus.
b- It is surrounded by a circular sulcus.
c- Its apex is referred to as the pulvinar.
d. Has three opercula, frontal, parietal and temporal.
**2- One of the opercula of the insula is: *(2020)*.**
a- Temporal.
b- Occipital.
c- Infundibular.
d- Para hippocampal.

**3- Insula is a part of the cerebral cortex hidden within the depth of: *(2019)*.**
a- Posterior ramus of lateral sulcus.
b- Central sulcus.
c- Superior temporal sulcus.
d- Intraparietal sulcus.
**4- One of the following gyri is not visible on the medial surface: *(2023)*.**
a- Lingual.
b- Callosal.
c- Cingulate.
d- Angular.
**5- One of the following sulci is on the medial surface of the cerebral hemisphere: (*2023)*.**
a- Cingulate sulcus.
b- Central sulcus.
c- Superior temporal sulcus.
d- Posterior ramus of lateral sulcus.
**6- Concerning the orbital surface of the cerebral hemisphere, one of the following is true: *(DEP BOOK - 2019)*.**
a- Is the anterior 1/3 of the inferior surface of the hemisphere.
b- The gyrus rectus lies lateral to the olfactory sulcus.
c- It rests on the tentorium cerebelli.
d- It presents the olfactory bulb and tract.
**7- Regarding the cingulate sulcus; choose one best answer: *(DEP BOOK)*.**
a- Begins above the rostrum of the corpus callosum.
b- Forms an arch about a finger breadth below the body of the corpus callosum.
c- Ends by dividing into two branches; submarginal and parietal.
d- Supplied by the middle cerebral artery.
**8- The cingulum: *(DEP BOOK)*.**
a- Connects the orbital gyri of the frontal lobe with the anterior part of the temporal lobe.
b- Makes arch over the stem of the lateral sulcus.
c- Is sharply bent on itself.
d- Supplied by the anterior cerebral artery.

VIP ACADEMY

## ***Base of brain***

**1- Which of the following is a part of the hypothalamus? *(2019)*.**
a- Pineal body.
b- Medial geniculate body.
c- Anterior commissure.
d- Mamillary bodies.
**2- One of the following structures is not a content of the interpeduncular fossa: *(2017)*.**
a- Mamillary bodies.
b- Optic tract.
c- Infundibulum.
d- Tuber cinereum.

**3- One of the following structures is a content of the interpeduncular fossa: *(2015)*.**
a- Lamina terminalis.
b- Posterior perforated substance.
c- Anterior perforated substance.
d- Olive.
e- Optic tract.
**4- One of the following is not a content of the interpeduncular fossa: *(2021 – IPKA - 2023)*.**
a- Tuber cinereum.
b- Mammillary bodies.
c- Posterior perforated substance.
d- Trochlear nerve.
**5- The interpeduncular fossa contain: *(2019)*.**
a- Anterior perforated substance.
b- Mammillary bodies.
c- Pineal stalk.
d- Tuber vermis.
**6- The cranial nerve that emerge from the interpeduncular fossa around medial aspect of the cerebral peduncle is: *(2023)*.**
a- 3rd.
b- 4th.
c- 5th.
d- 6th.
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**7- One of the following structures is a content of interpeduncular fossa: *(DEP BOOK)*.**
a- Mammillary bodies.
b- Tuberculum cinereum.
c- anterior perforated substance.
d- Optic tract.
**8- Concerning the interpeduncular fossa, one of the following is a boundary of the fossa: *(DEP BOOK)*.**
a- Cerebellar peduncles.
b- Optic tract.
c- Optic nerve.
d- Tuber cinereum.
**9- One of the following is correct: *(DEP BOOK)*.**
a- The anterior perforated substance Lies on the medial side of the optic chiasma.
b- The olfactory nerve filaments end in the olfactory bulb.
c- The lateral olfactory root ends in the cuneus.
d- The infundibulum connects the tuber cinereum with the parotid gland.

# ***Cerebral hemisphere white matter***

**1- One of the following does not belong to the long association fibers: *(2019)*.**
a- Fornix.
b- Superior longitudinal bundle.
c- Uncinate fasciculus.
d- Inferior longitudinal bundle.
**2- One of the following is a commissural fiber of the human cerebrum: *(2019 - 2023)*.**
a- Cingulum.
b- Uncinate fasciculus.
c- Superior longitudinal bundle.
d- Corpus callosum.
e- Inferior longitudinal bundle.
**3- Regarding the internal capsule, the auditory radiation passes in: *(2019 - 2018)*.**
a- The anterior limb.
b- The posterior limb.
c- The sublenticular part.
d- The retrolenticular part.
**4- One of the following arteries does not supply the internal capsule: *(2018)*.**
a- Posterior cerebral artery.
b- Anterior cerebral artery.
c- Middle cerebral artery.
d- Anterior choroidal artery.
**5- The greatest commissure in the brain is: *(2018)*.**
a- Habenular commissure.
b- Hippocampal commissure.
c- Corpus callosum.
d- Anterior commissure.
**6- The following are projection fibers of the cerebral hemisphere: *(2017)*.**
a- Corticobulbar.
b- Cingulum.
c- Arcuate.
d- Uncinate.
**7- A 47-year-old woman presents with signs of increased intracranial pressure (vomiting, headache, lethargy}. MRI shows a large tumor invading the head of the caudate nucleus, the rostral portion of the putamen and involving a fiber bundle located between these two structures. This fiber bundle is most likely the *(DEP BOOK)*.**
a- Anterior commissure.
b-Anterior limb of the internal capsule.
c- Column of the fornix.
d- External capsule.
e- Posterior limb of the internal capsule.

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**8- The following regarding the internal capsule is incorrect: *(2021 - IPKA)*.**
a- It is made up of grey matter.
b- The anterior limb separates the caudate nucleus from the lentiform nucleus.
c- Corticothalamic fibers are found in the anterior limb.
d-The posterior limb separates the thalamus from the lentiform nucleus.
**9- Regarding corpus callosum one Is wrong: *(2021 - IPKA)*.**
a- Its body forms the roof of the trunk (body) of lateral ventricle.
b- Its upper surface is related to the anterior cerebral artery.
c- Splenium is supplied by the anterior cerebral artery.
d- The fibers in the genu forms forceps minor.
**10- The corticobulbar fibers pass through the following part of the internal capsule: *(2021 – 2019 - 2020)*.**
a- Anterior limb.
b- Genu.
c- Posterior limb.
d- Retrolenticular part.
**11- One of the following is not association fibers: *(2020)*.**
a- Corpus callosum.
b- Superior longitudinal bundle.
c- Inferior longitudinal bundle.
d- Cingulum.
**12- One of the following is not an association fiber of the Cerebrum: *(2021 - 2023)*.**
a- Uncinate fasiculus.
b- Cingulum.
c- Corona radiata.
d- Superior longitudinal bundle.
**13- In which part of the internal capsule does the auditory radiation pass: *(2020 - 2019)*.**
a- Anterior limb.
b- Retrolentiform.
c- Sublentiform.
d- Genu.
**14- The commissural fibers of the brain include all of the following except: *(2019)*.**
a- Corpus callosum.
b- Cingulum.
c- Anterior commissure.
d- Hippocampal commissure.
**15- One of the following, belongs to the projection fibers: *(2019)*.**
a- Internal capsule.
b- Superior longitudinal bundle.
c- Uncinate fasciculus.
d- Inferior longitudinal bundle.

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**16- The part of the internal capsule that contains optic radiation is: *(2024)*.**
a- Anterior limb.
b- Posterior limb.
c- Retro-lenticular part.
d- Sub-lenticular part.
**17- Fibers connecting corresponding areas in the 2 cerebral hemispheres: *(2024)*.**
a- Projection fibers .
b- Corpus callosum .
c- Cingulum.
d- Superior longitudinal fibers.
**18- Lesion of the internal capsule results in one of the following: *(DEP BOOK)*.**
a- Monoplegia on the same side.
b- Monoplegia on the opposite side.
c- Hemiplegia on the same side.
d- Hemiplegia on the opposite side.
**19- The anterior commissure mainly connects the right and left: *(DEP BOOK)*.**
a- Temporal poles.
b- Frontal poles.
c- Occipital poles.
d- Olfactory bulbs and tracts.
**20- One of the followings structures belongs to the visual pathway: *(DEP BOOK)*.**
a- Sublenticular part of internal capsule.
b- Retrolenticular part of internal capsule.
c- Inferior colliculus.
d- Medial geniculate body.
**21- The pyramidal tract fibers pass through the following parts of internal capsule: *(DEP BOOK)*.**
a- Anterior and posterior limbs.
b- Anterior limb and genu.
c- Posterior limb and genu.
d- Retrolenticular and genu.
**22- The corticobulbar fibers pass through the following part of internal capsule: *(DEP BOOK)*.**
a- Anterior limb.
b- Genu.
c- Posterior limb.
d- Retrolentiform.
**23- The projection fibers of the cerebral hemisphere: *(DEP BOOK)*.**
a- Contain fibers from the thalamus to the cerebellar cortex.
b- Connect same areas in the different hemisphere.
c- All its fibers pass in the external capsule.
d- Connect different areas in the same hemisphere.

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**24- One of the following, belongs to the projection fibers: *(DEP BOOK)*.**
a- Internal capsule.
b- Superior longitudinal bundle.
c- Uncinate fasciculus.
d- Cingulum.

## ***Basal ganglia***

**1- Which of the following structures has an extrapyramidal function: *(2017)*.**
a- Amygdaloid nucleus.
b- Claustrurn.
c- Corpus striatum.
d- Fornix.
**2- A 48-year-old man presents with a movement disorder (chorea} and mental deterioration. MRI shows the loss of a structure in the wall of the anterior horn of the lateral ventricle. Which of following is most likely lost in this patient? *(DEP BOOK)*.**
a- Anterior thalamic nucleus.
b- Body of the caudate nucleus.
c- Column of the fornix.
d- Amygdala.
e- Head of the caudate nucleus.
**3- The basal ganglia include all the following except: *(2020 - 2019)*.**
a- Caudate nucleus.
b- Lentiform nucleus.
c- Claustrum.
d- Fornix.
**4- The structure separates the lentiform nucleus from the claustrum is: *(2023)*.**
a- Thalamus.
b- External capsule.
c- Internal capsule.
d- Head of caudate.
**5- Which of the followings concerning the basal ganglia is correct: *(2023)*.**
a- The amygdaloid nucleus is connected to the caudate.
b- The claustrum is not a part of basal ganglia.
c- The corpus striatum lies medial to thalamus.
d- The lentiform nucleus is completely divided by external capsule into globus pallidus and putamen.

VIP ACADEMY

# ***Lateral ventricle***

**1- Concerning the lateral ventricle, choose one correct answer: *(2019)*.**
a- Its anterior horn lies in the parietal lobe.
b- Its posterior horn lies in the temporal Jobe.
c- Its the cavity of the cerebral hemisphere.
d- Has no choroid plexus.
**2- Regarding the anterior horn of lateral ventricle, choose the correct answer: *(2019)*.**
a- Bounded anteriorly by pineal body.
b- The floor is formed by rostrum of corpus callosum.
c- It is present in the parietal lobe of the cerebrum.
d- It is related posteriorly to posterior commissure.
**3- One of the following structures lies in the roof of inferior horn of lateral ventricle: *(2021 - IPKA)*.**
a- Hippocampus.
b- Fimbria.
c- Collateral eminence.
d- Tail of the caudate nucleus.
**4- Concerning the lateral ventricle, select the incorrect statement: *(2021)*.**
a- Its body ends opposite the splenium of corpus callosum.
b- The lateral wall of anterior horn is related to the body of the fornix.
c- The septum pellucidum lies in the medial wall of the body.
d- The inferior horn extends forwards in the temporal lobe.
**5- The lateral ventricle communicates with the 3rd ventricle through: *(2019)*.**
a- Foramen of Monro.
b- Fora men of Magendie.
c- Foramen of Luschka.
d- Aqueduct of Sylvius.
**6- The lateral ventricles are located within which part of the brain? *(2024)*.**
a- Cerebellum.
b- Brainstem.
c- Cerebral hemispheres.
d- Diencephalon.
**7- Concerning the lateral ventricle, choose one correct answer: *(DEP BOOK)*.**
a- Its anterior horn lies in the parietal lobe.
b- Its posterior horn lies in the temporal lobe.
c- Is the only cavity in the telencephalon.
d- Has no choroid plexus.
**8- Relations of the anterior horn of lateral ventricle: *(DEP BOOK)*.**
a- The roof is formed by head of caudate nucleus.
b- Lateral wall is formed by anterior part of septum pellucidum.
c- Floor is formed by body of corpus callosum.
d- Anteriorly is limited by genu of corpus callosum.

VIP ACADEMY

# ***Diencephalon***

**1- The lateral surface of diencephalon is related to one of the following structures: *(2018)*.**
a- Third ventricle.
b- Fourth ventricle.
c- Claustrum.
d- Internal capsule.
**2- Which of the following is not part of the diencephalon: (*DEP BOOK*).**
a- Mammillary bodies.
b- Tuber cinereum.
c- Anterior perforated substance .
d- Subthalamus.
**3- Which one of the following structures is found in the diencephalon: *(2020)*.**
a- Internal capsule.
b- Caudate nucleus.
c- Globus pallidus.
d- Medial and lateral geniculate bodies.
**4- The lateral surface of the thalamus is related directly to one of the following structures: *(2020 - 2019)*.**
a- Third ventricle.
b- Claustrum.
c- External capsule.
d- Posterior limb of internal capsule.
**5- The mammillary bodies are part of *(2023)*.**
a- Pons.
b- Midbrain.
c- Diencephalon.
d- Medulla.
**6- The part of the diencephalon attached to the posterior part of the thalamus: *(2024)*.**
a- Hypothalamus.
b- Epithalamus.
c- Metathalamus.
d- Subthalamus.
**7- Which of the following is part of the diencephalon: *(DEP BOOK)*.**
a- Mammillary bodies.
b- Tuberculum cinereum.
c- Anterior perforated substance.
d- Optic tract.
**8- Which of the following is a part of the hypothalamus: *(DEP BOOK)*.**
a- Optic chiasma.
b- Infundibulum.
c- Tuberculum cinereum.
d- Anterior commissure.

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**9- The lateral surface of diencephalon is related to one of the following structures: *(DEP BOOK)*.**
a- Third ventricle.
b- Claustrum.
c- External capsule.
d- Internal capsule.
**10- Which of the following is a part of the hypothalamus? *(2019)*.**
a- Pineal body.
b- Medial geniculate body.
c- Anterior commissure.
d- Mamillary bodies.

## ***Third ventricle***

**1-The following statements concern the third ventricle, one of them is correct: *(DEP BOOK)*.**
a- It is the cavity of the hindbrain.
b- It communicates with the lateral ventricles through the cerebral aqueduct.
c- It is continuous with fourth ventricle through the interventricular foramen.
d- The choroid plexus is located in the roof .
e- The choroid plexus receives its arterial supply through the posterior inferior cerebellar arteries.
**2- The third ventricle, choose the false statement: *(2021 - IPKA)*.**
a- Communicates with the 4$^{th}$ ventricle through the cerebral aqueduct.
b- The corpus callosum is a direct superior relation.
c- Has a recess related to the optic chiasma.
d- Is bounded anteriorly by the lamina terminalis.
**3- Pineal gland forms part of the: *(2020)*.**
a- Anterior wall of 3$^{rd}$ ventricle.
b- Anterior wall of 4$^{th}$ ventricle.
c- Posterior wall of 3$^{rd}$ ventricle.
d- Floor of 3$^{rd}$ ventricle.
**4- Which part of the ventricular system contains the choroid plexus: *(2020)*.**
a- Frontal horn of lateral ventricle.
b- Occipital horn of lateral ventricle.
c- Roof of third ventricle.
d- Cerebral aqueduct.
**5- The anterior wall of 3rd ventricle is formed by all of the following except: *(2019)*.**
a- Anterior commissure.
b- Anterior column of the fornix.
c- Lamina terminalis.
d- Septum pellucidum.
**6- The lateral ventricle communicates with the 3$^{rd}$ ventricle through: *(2019 - 2023)*.**
a- Foramen of Monro.
b- Fora men of Magendie.
c- Foramen of Luschka.
d- Aqueduct of Sylvius.

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**7- The following are recesses of 3rd ventricle except: *(2019)*.**
a- Pineal recess.
b- Optic recess.
c- Infundibular recess.
d- Lateral recess.
**8- The 3rd ventricle is bounded anteriorly by: *(2023)*.**
a- Pineal body.
b- Subthalamus.
c- Lamina terminalis.
d- Habenular commissure.
**9- The following statements concern the third ventricle, one of them is correct: *(2024)*.**
a- It is the cavity of the hindbrain.
b- It communicates with the lateral ventricles through the cerebral aqueduct.
c- It is continuous with the fourth ventricle through the interventricular foramen.
d- The choroid plexus is located in the roof.
**10- The following statements concern the third ventricle, one of them is correct: *(DEP BOOK)*.**
a- It is the cavity of the hindbrain.
b- It communicates with the lateral ventricles through the cerebral aqueduct.
c- It is continuous with the fourth ventricle through the interventricular foramen.
d- The choroid plexus is located in the roof.

VIP ACADEMY

## ***Meninges***

**1- The cisterna magna of subarachnoid space is a name given to the: *(2019)*.**
a- Interpeduncular cistern.
b- Cistern of the corpus callosum.
c- Cistern of the lateral fissure.
d- Cerebello-medullary cistern.
**2- Which statement about the meninges is most accurate? *(DEP BOOK)*.**
a- Dura mater has two distinct layers surrounding brain and spinal cord.
b- The major venous sinuses lie in the subdural space.
c- Rupture of middle meningeal artery causes subarachnoid hemorrhage.
d- Dura mater is thickened either side of the spinal cord to form the denticulate ligaments.
e- Pia mater pierces the distal extreme of the dura as the filum terminale.
**3- The cisterna magna of the subarachnoid space is a name given to the: *(2021 - 2024)*.**
a- Cerebello-medullary cistern.
b- Interpeduncular cistern.
c- Cistern of the lateral fissure.
d- Pontine cistern.
**4- Which part of the ventricular system contains the choroid plexus: *(2020)*.**
a- Frontal horn of lateral ventricle.
b- Occipital horn of lateral ventricle.
c- Roof of third ventricle.
d- Cerebral aqueduct.

**5- Regarding cranial meninges, one of the following is true: *(2019)*.**
a- Outer layer of the dura forms dural folds to separate different parts of brain.
b- Subarachnoid space contains CSF.
c- Extradural space is the space between the dura and arachnoid.
d- The dural fold between cerebral and cerebellar hemispheres is called falx cerebelli.
**6- Pia matter: *(2024)*.**
a- Tough membrane.
b- Is carried into all its sulci.
c- Bridges over the various sulci on the surface of the brain.
d- Its two layers enclose venous sinuses.
**7- The cisterna magna of subarachnoid space is a name given to the: *(DEP BOOK)*.**
a- Interpeduncular cistern.
b- Cistern of the corpus callosum.
c- Cistern of the lateral fissure.
d- Cerebellomedullary cistern.
**8- Regarding meninges, one of the following is true: *(DEP BOOK)*.**
a- Inner layer of dura forms folds to separate different parts of brain.
b- Extradural space is between dura & arachnoid.
c- dura dips into the sulci of brains.
d- Subarachnoid space contains CSF.
**9- Concerning the arachnoid villi and granulations; choose one best answer: *(DEP BOOK)*.**
a- Never indent the bone.
b- They are found in large numbers along the superior sagittal sinus.
c- Increase in size in children.
d- Help in formation of CSF.

VIP ACADEMY

## ***Cerebrospinal fluid (CSF)***

**1- A 9-month-old infant presented with hydrocephalus (intracranial accumulation of CSF). What is the possible site of obstruction? *(2018 - 2016)*.**
a- Foramina of Magendie and Luschka.
b- Foramen caecum.
c- Central canal of spinal cord.
d- 5th ventricle.
**2- The sites where cerebrospinal fluid passes from around the brain into the venous system is/are called: *(2019)*.**
a- Choroid plexuses.
b- Lateral apertures.
c- Median aperture.
d- Arachnoid villi.

**3- A 55-year-old male farmer was admitted to the emergency department after falling. from hayloft. Radiographic examination revealed a small, depressed fracture of the. skull vertex and thrombosis of the superior sagittal sinus. A day later, the patient lost. his consciousness. What is the most likely cause for his loss of consciousness? *(2018)*.**

a- Obstruction of CSF resorption.
b- Obstruction of the cerebral aqueduct of Sylvius.
c- Laceration of the middle meningeal artery.
d- Aneurysm of the middle cerebral artery.

**4- A 5-day-old infant was diagnosed with a noncommunicating hydrocephalus (intracranial accumulation of CSF). Which of the following is most likely to lead to such a condition? *(2015)*.**

a- Obstruction in the circulation of the cerebrospinal fluid.
b- Increased size of the head.
c- Disturbances in the resorption of cerebrospinal fluid.
d- Excess production of cerebrospinal fluid.
e- Failure of the neural tube to close.

**5- Regarding the ventricular system and cerebrospinal fluid (CSF) production, which of the following statements is true? *(DEP BOOK)*.**

a- CSF is exclusively produced in the choroid plexi of the lateral ventricles.
b- CSF passes from the 3rd to the 4th ventricle via the interventricular foramen.
c- The 4th ventricle communicates directly with the subarachnoid space.
d- The total daily production of CSF is approximately 150 ml.
e- CSF is reabsorbed mainly through the arachnoid granulations in the. inferior sagittal sinus.

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**6- Foramen of Luschka connects the: *(2021)*.**

a- Lateral ventricle with the third ventricle.
b- Two lateral ventricles.
c- Third ventricle with the fourth ventricle.
d- Fourth ventricle with the subarachnoid space.

**7- One of the following is not true concerning the arachnoid granulations: *(2021)*.**

a- They are found along the superior sagittal sinus.
b- They produce C.S.F.
c- They consist of aggregation of arachnoid villi.
d- They indent the overlying skull bone in the old age.

**8- Cerebrospinal fluid circulates through all of the following except: *(2019)*.**

a- Corpus callosum.
b- Lateral ventricles.
c- Fourth ventricle.
d- Subarachnoid space.

**9- The CSF is produced by choroid plexus present in: *(2019)*.**

a- Roof of 3rd ventricle.
b- Floor of 3rd ventricle.
c- Floor of posterior horn of lateral ventricle.
d- Lateral wall of inferior horn of lateral ventricle.

**10- Arachnoid Villi: *(2023)*.**
a- Are situated in the spinal cord.
b- Produce CSF.
c- Pass CSF to venous system.
d- Are part of dura.
**11- Concerning the cerebrospinal fluid, choose one correct answer: *(DEP BOOK)*.**
a- Produced mainly in the subarachnoid space.
b- Produced by the arachnoid granulations.
c- Third ventricle has a choroid plexus in its roof.
d- Fourth ventricle has a choroid plexus in its floor.
**12- Concerning the choroid plexus; choose one best answer: *(DEP BOOK)*.**
a- All ventricles have choroids plexuses.
b- Is formed by a plexus of capillaries and a thin layer of cuboidal epithelium.
c- Is responsible for absorption of the cerebrospinal fluid.
d- All of the choroids plexuses are present in the floor of the ventricles.

## ***Arterial and venous supply of brain***

**1- One of the following arteries is a branch of the basilar artery: *(2019)*.**
a- Anterior inferior cerebellar artery.
b-Posterior inferior cerebellar artery.
c- Anterior spinal artery.
d- Posterior spinal artery.
**2- One of the following arteries does not supply the internal capsule: *(2018 - 2019)*.**
a- Posterior cerebral artery.
b- Anterior cerebral artery.
c- Middle cerebral artery.
d- Anterior choroidal artery.
**3- The main blood supply of the medial surface of the cerebrum is: *(2018)*.**
a- Posterior inferior cerebellar.
b- Anterior cerebral.
c- Middle cerebral.
d- Posterior cerebral.
**4- Injury of which of the following arteries results in lateral medullary syndrome: (*2017*).**
a- Superior cerebellar.
b- Posterior inferior cerebellar.
c- Anterior inferior cerebellar.
d- Posterior cerebral.
**5- The choroid plexus of the 4th ventricle is derived from the following artery: *(2017)*.**
a- Posterior inferior cerebellar.
b- Anterior inferior cerebellar.
c- Superior cerebellar.
d- Middle cerebral.

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**6- A 65-year-old patient was admitted to the hospital in an unconscious condition. A CT scan examination revealed that the patient had suffered a cerebral vascular accident (stroke), with a small hematoma produced by the superior cerebellar artery. Which of the following nerves is most likely affected by the hematoma? *(2017)*.**
a- Trochlear.
b- Abducent.
c- Facial.
d- Vestibulocochlear.
**7- The following artery shares in the supply of the medulla: *(2016)*.**
a- Posterior inferior cerebellar.
b- Superior cerebellar.
c- Posterior cerebral.
d- Middle cerebral.
**8- Branches of which cerebral artery vascularize the primary auditory cortex, Wernick's area and angular gyrus: *(DEP BOOK)*.**
a- Thalamogeniculate.
b- Anterior choroidal.
c- Middle cerebral.
d- Anterior cerebral.
**9- What is the major vessel lateral to optic chiasma: *(DEP BOOK)*.**
a- Anterior cerebral artery.
b- Posterior cerebral.
c- Posterior communicating.
d- Posterior inferior cerebellar artery.
**10- Anterior cerebral artery supplies: *(DEP BOOK)*.**
a- Paracentral lobule.
b- Insula.
c- Visual areas.
d- Wernick's area.
**11- Regarding the anterior cerebral artery, select the false statement: *(2021 - IPKA)*.**
a- Is a branch arises from the internal carotid artery.
b- Runs backwards along the upper surface of corpus callosum.
c- Has no central branches.
d- Supplies all parts of the corpus callosum, except the splenium.
**12- One of the following is a branch of the basilar artery: *(2021 - 2023)*.**
a- Anterior inferior cerebellar artery.
b- Posterior inferior cerebellar artery.
c- Anterior spinal artery.
d- Posterior spinal artery.
e- Anterior cerebral artery.
f- Middle cerebral artery.

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**13- One of the following arteries is not part of the circulus arteriosus: *(2021 - IPKA)*.**
a- Anterior communicating artery.
b- Posterior communicating artery.
c- Anterior cerebral artery.
d- Middle cerebral artery.
**14- One of the following is a branch of the vertebral artery: *(2020)*.**
a- Anterior cerebral artery.
b- Anterior inferior cerebellar artery (AICA).
c- Posterior inferior cerebellar artery (PICA).
d- Superior cerebellar artery.
**15- The middle cerebral artery supplies one of the following parts: *(2020 – 2019 - 2024)*.**
a- The retro-lentiform part of internal capsule.
b- Motor and sensory areas of the whole body.
c- Medial half of the orbital surface of the cerebral hemisphere.
d- Auditory area.
e- The sublentiform part of internal capsule.
**16- Which one of the following vessels does not contribute to the circle of Willies: (2020*)*.**
a- Middle cerebral artery.
b- Internal carotid artery.
c- Anterior communicating.
d- Posterior cerebral artery.
**17- Which of the following arteries supply the Brocas area: *(2023)*.**
a- Anterior cerebral artery.
b- Middle cerebral artery.
c- Posterior cerebral artery.
d- Vertebral artery.
**18- Concerning the cerebellum, select the true statement: *(2023)*.**
a- It lies in the middle cranial fossa.
b- Its upper surface is covered by falx cerebri.
c- Its superior vermis has a nodule and uvula only.
d- It's supplied by superior cerebellar artery.
**19- Concerning the superficial middle cerebral vein choose one best answer: *(2024)*.**
a- It lies in the posterior ramus of the lateral sulcus.
b- It receives the majority of inferior cerebral veins of the medial surface.
c- Ends in the transverse sinus.
d- Ends in the superior sagittal sinus.
**20- Regarding the arterial supply of the brain, one of the following is true: *(DEP BOOK)*.**
a- The areas of smell and vision are supplied by the middle cerebral artery.
b- The cerebellum is supplied by branches of the internal carotid system.
c- The anterior inferior cerebellar artery shares in the arterial supply of the medulla.
d- The auditory and speech areas are supplied by branches of the middle cerebral artery.

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**21- Concerning the superficial middle cerebral vein choose one best answer: *(DEP BOOK)*.**
a- It lies in the posterior ramus of the lateral sulcus.
b- It receives the majority of inferior cerebral veins of the medial surface.
c- Ends in the transverse sinus.
d- Ends in the superior sagittal sinus.
**22- The anterior cerebral artery supplies: *(DEP BOOK)*.**
a- The motor and sensory areas of the upper and lower limbs.
b- The occipital lobe of the brain.
c- All parts of the internal capsule.
d- All parts of the corpus callosum, except splenium.
**23- The posterior cerebral artery supplies one of the following structures: *(DEP BOOK)*.**
a- Splenium of corpus callosum.
b- Insula.
c- Lateral half of orbital surface of cerebral hemisphere.
d- Main part of the lateral surface of the hemisphere.
**24- The middle cerebral artery supplies one of the following parts: *(DEP BOOK)*.**
a- Auditory area.
b- The sublentiform part of internal capsule.
c- Motor and sensory areas of the whole body.
d- Medial half of orbital surface of cerebral hemisphere.
**25- Branches of which cerebral artery vascularize the primary auditory cortex, Wernick's area and angular gyrus: *(DEP BOOK)*.**
a- Thalamogeniculate.
b- Anterior cerebral artery.
c- Middle cerebral artery.
d- Anterior choroidal artery.

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## ***Development of CNS***

**1- The neural tube gives rise to which of the following cells: *(2018)*.**
a- Cells of the sympathetic trunk.
b- Cells of the suprarenal medulla.
c- Dorsal root ganglion cells.
d- Ventral horn cells.
**2- Infundibulum is a downward extension of the: *(2016)*.**
a- Prosencephalon.
b- Telencephalon.
c- Rhombencephalon.
d- Diencephalon.
**3- Of the following, the one most closely associated with the cerebrum is:*(DEP BOOK)*.**
a- Telencephalon.
b- Diencephalon.
c- Mesencephalon.
d- Metencephalon.

**4- Of the following, the one most closely associated with the 3rd ventricle is: *(DEP BOOK)*.**
a- Telencephalon.
b- Diencephalon.
c- Mesencephalon.
d- Metencephalon.
**5- Which spinal cord layer contains cell bodies and astrocytes? *(DEP BOOK)*.**
a- Ventricular.
b- Ependyma.
c- Mantle.
d- Marginal.
**6- The brain flexure which develops between the metencephalon and the myelencephalon is called the: *(DEP BOOK)*.**
a- Midbrain.
b- Hindbrain.
c- Cervical.
d- Cephalic.
e- Pontine.
**7- Regarding the development of the spinal cord, the following are correct statement except: *(2020)*.**
a- White matter of the spinal cord develops in the marginal zone.
b- Dorsal root ganglia are derived from the neural crest cells.
c- At birth spinal cord ends at the level of the first lumbar vertebra.
d- Mantle zone gives rise to the grey matter of the spinal cord.
**8- Regarding the development of the central nervous system, mark one correct statement: *(DEP BOOK)***

a- It appears as a thickened ectodermal neural plate

b- Prosencephalon gives rise to metencephalon and diencephalon

c- Cervical flexure appears at the junction of hindbrain and spinal cord

d- Myelencephalon will form pons and cerebellum

**9- Regarding the development of the central nervous system, mark one correct statement: *(DEP BOOK)***

a- The cephalic part of the neural tube remains narrow

b- Its neuropores establish a temporary connection of the neural tube with the amnion

c- The telencephalon forms pons and cerebellum

d- The myelencephalon forms midbrain

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**10- Regarding the development of the brain, mark one correct statement: *(DEP BOOK)***

a- The telencephalon gives rise to thalamus and hypothalamus

b- Metencephalon gives rise to medulla oblongata

c- The cavity of the rhombencephalon is known as fourth ventricle

d- The cavity of the mesencephalon is known as lateral ventricle

**11- Regarding the development of the spinal cord, mark one correct answer: *(DEP BOOK)***

a- Grey matter of the spinal cord develops in the marginal zone

b- Dorsal root ganglia are derived from the neural crest cells

c- At birth spinal cord ends at the level of the first lumbar vertebra

d- Mantle zone gives rise to the white matter of the spinal cord

**12- Regarding the development of the spinal cord, mark one correct answer: *(DEP BOOK)***

a- Dorsal root ganglia are derived from the neural tube

b- At birth spinal cord ends at the level of the second lumbar vertebra

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c- Spina bifida occulta is a condition in which there is failure of the lamina to fuse in three successive vertebrae

d- Meningocele is the condition in which spina bifida is associated with herniation of meninges

**13- Regarding the development of the spinal cord, mark one correct answer: *(DEP BOOK)*.**

a- Basal plate forms the sensory areas of the spinal cord

b- Alar plate forms the motor areas of the spinal cord

c- Sulcus limitans marks the boundary between motor and sensory areas

d- The roof and floor plates are thick

# Model answer

## ***Spinal cord***

| | | | | | | |
|---|---|---|---|---|---|---|
| 1-a | 2-c | 3-b | 4-b | 5-b | 6-c | 7-c |
| 8-c | 9-b | 10-c | 11-d | 12-b | 13-d | 14-d |
| 15-a | 16-d | 17-a | 18-c | 19-d | 20-c | 21-b |
| 22-c | 23-c | 24-b | 25-d | 26-a | 27-b | 28-d |
| 29-c | 30-c | | | | | |

## ***Brain stem***

| | | | | | | |
|---|---|---|---|---|---|---|
| 1-c | 2-b | 3-c | 4-d | 5-c | 6-a | 7-d |
| 8-b | 9-d | 10-a | 11-c | 12-d | 13-a | 14-a |
| 15-b | 16-a | 17-d | 18-a | 19-c | 20-c | 21-c |
| 22-c | 23-d | 24-d | 25-a | 26-a | 27-a | 28-d |
| 29-d | 30-c | 31-d | 32-a | 33-b | 34-b | 35-c |
| 36-c | 37-d | 38-a | 39-b | 40-a | 41-b | 42-c |
| 43-b | 44-d | 45-c | 46-b | 47-d | 48-b | |

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## ***Cerebellum***

| | | | | | | |
|---|---|---|---|---|---|---|
| 1-b | 2-d | 3-c | 4-d | 5-b | 6-d | 7-c |
| 8-c | 9-d | 10-c | 11-b | 12-a | 13-c | 14-d |
| 15-b | 16-a | 17-a | | | | |

## ***Fourth ventricle***

| | | | | | | |
|---|---|---|---|---|---|---|
| 1-d | 2-a | 3-d | 4-d | 5-b | 6-d | 7-b |
| 8-a | 9-b | 10-a | 11-c | 12-b | 13-c | |

## ***Sulci & gyri***

| | | | | | | |
|---|---|---|---|---|---|---|
| 1-c | 2-a | 3-a | 4-d | 5-a | 6-d | 7-c |
| 8-d | | | | | | |

## ***Base of brain***

| | | | | | | |
|---|---|---|---|---|---|---|
| 1-d | 2-b | 3-b | 4-d | 5-b | 6-a | 7-a |

| | | | | | | |
|---|---|---|---|---|---|---|
| 8-b | 9-b | | | | | |

## ***Cerebral hemisphere white matter***

| | | | | | | |
|---|---|---|---|---|---|---|
| 1-a | 2-d | 3-c | 4-a | 5-c | 6-a | 7-b |
| 8-a | 9-c | 10-b | 11-a | 12-c | 13-c | 14-b |
| 15-a | 16-c | 17-b | 18-d | 19-d | 20-b | 21-c |
| 22-b | 23-a | 24-a | | | | |

## ***Basal ganglia***

| | | | | | | |
|---|---|---|---|---|---|---|
| 1-c | 2-e | 3-d | 4-b | 5-a | | |

## ***Lateral ventricle***

| | | | | | | |
|---|---|---|---|---|---|---|
| 1-c | 2-b | 3-d | 4-b | 5-a | 6-c | 7-c |
| 8-d | | | | | | |

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## ***Diencephalon***

| | | | | | | |
|---|---|---|---|---|---|---|
| 1-d | 2-c | 3-d | 4-d | 5-c | 6-c | 7-a |
| 8-b | 9-d | 10-d | | | | |

## ***Third ventricle***

| | | | | | | |
|---|---|---|---|---|---|---|
| 1-d | 2-b | 3-c | 4-c | 5-d | 6-a | 7-d |
| 8-c | 9-d | 10-d | | | | |

## ***Meninges***

| | | | | | | |
|---|---|---|---|---|---|---|
| 1-d | 2-e | 3-a | 4-c | 5-b | 6-b | 7-d |
| 8-d | 9-b | | | | | |

## ***Cerebrospinal fluid (CSF)***

| | | | | | | |
|---|---|---|---|---|---|---|
| 1-a | 2-d | 3-a | 4-a | 5-a | 6-d | 7-b |
| 8-a | 9-a | 10-c | 11-c | 12-b | | |

## ***Arterial and venous supply of brain***

| | | | | | | |
|---|---|---|---|---|---|---|
| 1-a | 2-a | 3-b | 4-b | 5-a | 6-a | 7-a |
| 8-c | 9-a | 10-a | 11-c | 12-a | 13-d | 14-c |
| 15-d | 16-a | 17-b | 18-d | 19-a | 20-d | 21-a |
| 22-d | 23-a | 24-a | 25-c | | | |

## ***Development of CNS***

| | | | | | | |
|---|---|---|---|---|---|---|
| 1-d | 2-d | 3-a | 4-b | 5-c | 6-e | 7-c |
| 8-a & c | 9-b | 10-c | 11-b | 12-d | 13-c | |

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